

CRONICĂ

Party Platform & Comic AI Game

Game Design Document · v0.2 · Confidential

Platform Codename: **AGORA**
First Game Codename: **CRONICĂ**

Romanian-first · 18+ · LAN/Discord · AI-powered

Prepared by: Technical Lead
Status: Pre-production · Pending approval to begin M1

Executive Summary

CRONICĂ is an AI-powered party game and the flagship title of AGORA — a reusable, extensible party game platform designed to become the Romanian equivalent of Jackbox Games. It is not a Jackbox clone. It is a new category of party game experience built around AI-generated content that feels genuinely surprising, funny, and unrepeatable.

Players submit answers to absurd prompts on their phones. The AI pipeline — guided by a Creative Director — generates an original comic story in one of several cinematic genres. The comic is then revealed panel by panel with narration, sound effects, and cinematic animation. Players vote. The best comic wins. The cycle repeats.

The platform (AGORA) handles all shared infrastructure: lobbying, networking, player management, timers, voting, and scoring. CRONICĂ is Game #1 built on top of it. Future games can be added to the platform without re-engineering the foundation.

Attribute	Value
Platform name	AGORA
Game #1 name	CRONICĂ
Target audience	18+, friends, LAN parties, Discord calls, streamers
Language	Romanian first, English later
Deployment	Local Windows PC (RTX 4070), phones via browser, no install
AI stack	Ollama + Llama 3.1 8B, ComfyUI + FLUX.1 schnell, ElevenLabs TTS (Piper offline fallback)
Tech stack	Node.js 22 LTS, Fastify, Socket.IO, Svelte 5, Tauri 2, Python 3.12
Licencing model	TBD — architecture supports commercial from day one
Current status	Pre-production, GDD approval pending

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This document covers: Platform Vision · Game Design · Player Experience · AI Systems · Creative Director · Technical Architecture · Risk Analysis · Milestones

1 Platform Vision: AGORA

1.1 Why a Platform, Not a Single Game

The most expensive part of any party game is the shared infrastructure: lobbying, QR code joining, phone clients, timers, round management, voting, leaderboards, and asset delivery. Building this once and reusing it across multiple games is the correct architectural decision.

AGORA is the engine. CRONICĂ is the first game. Future titles — quiz games, drawing games, debate games, improv games — all inherit AGORA without re-engineering networking or phone clients.

DECISION AGORA must be designed so that adding a new game requires only: a new game module + new phone UI screens + new asset pipeline. Zero changes to core infrastructure.

1.2 Platform Components

Component	Responsibility	Reused by all games?
Lobby Manager	Room creation, QR code, player join/leave, host controls	Yes
Player Manager	Nicknames, avatars, connection state, reconnection	Yes
Round Engine	Phase sequencing, timers, state machine	Yes
Submission Collector	Receives and stores player answers per phase	Yes
Voting Engine	Distributes options, collects votes, resolves ties	Yes
Scoring Engine	Points, streaks, bonuses, round/game totals	Yes
Asset Delivery	Serves generated images and audio to presenter	Yes
AI Pipeline	Story → images → audio orchestration	Per game (pluggable)
Presenter Layer	Cinematic reveal, animation, effects	Per game
Phone Client Shell	QR join, name entry, connection indicator	Yes
Phone Game Screens	Answer prompts, vote, react, wait screen	Per game

1.3 Platform Extensibility Contract

Every game built on AGORA must implement exactly one interface: the Game Module Interface. This is the contract between the platform and the game.

```
interface AgoraGameModule {  
  id:      string           // unique game identifier  
  name:    string           // display name  
  minPlayers: number  
  maxPlayers: number  
  phases:  PhaseDefinition[] // ordered list of round phases  
  promptPack: PromptPack    // questions/prompts for this game  
  pipeline: AIPipeline       // pluggable AI pipeline (optional)  
  presenterUI: PresenterModule // Tauri UI for this game  
  phoneUI:  PhoneModule      // Svelte screens for this game  
}
```

CRONICĂ implements this interface. A future quiz game implements it differently. AGORA does not care what the game does — only that it honours the interface.

2 Game Design: CRONICĂ

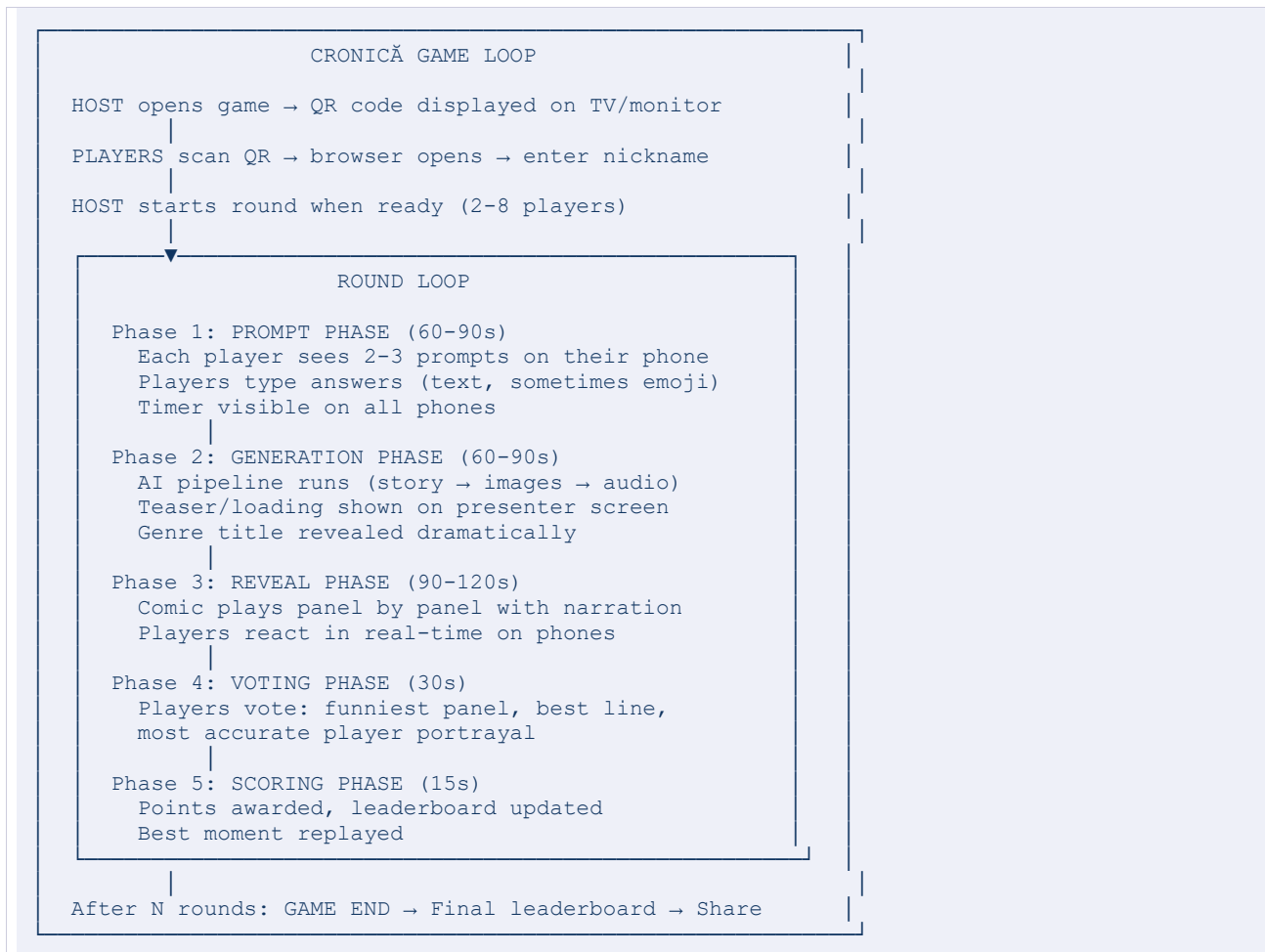
2.1 Core Concept

Players answer absurd, funny, or deeply personal prompts. The AI reads all answers and writes an original Romanian story in a randomly selected cinematic genre. It turns that story into a comic with generated art, narration, sound, and cinematic animation. The comic is revealed to everyone simultaneously. Players vote on which element was funniest. Points are awarded. Repeat.

The reveal is the reward. The comic is the reward. The narrator is the reward. Players should feel genuine surprise at what the AI created from their inputs.

DESIGN PRINCIPLE The AI must feel like a writer who happened to overhear the players, not like a machine that filled blanks in a template. Player answers are fuel, not slots.

2.2 Core Gameplay Loop



2.3 Player Count & Session Length

Players	Recommended Rounds	Estimated Session	Notes
2–3	5–6	45–60 min	Intimate; prompts adjusted for fewer inputs
4–6	4–5	60–75 min	Sweet spot; richest story inputs
7–8	3–4	60–90 min	Max chaos; funniest reveals
8+	Not recommended	—	Story coherence degrades with too many inputs

3 Player Experience

3.1 Journey Map: From Zero to First Laugh

Moment	What Happens	Emotion Target	Design Notes
Host opens app	QR code + room code appear on TV	Curiosity, anticipation	QR must be large, scannable from 3m away
Player scans QR	Phone opens instantly, no install	Relief, delight	Page must load in <2s on any phone browser
Enter nickname	Name appears on TV in real time	Social recognition	TV shows each new player joining with a sound
Prompts appear	Personal, specific, funny prompts	Mild panic, excitement	Players should feel slightly exposed
Submit answers	Confirmation + wait screen	Suspense	Show how many players still typing
Generation starts	Dramatic genre reveal on TV	Excitement	Genre title should feel like a film title card
Comic begins	First panel appears with narration	Shock, laughter	First panel must land a joke or reveal
Recognition moment	Player sees their answer in story	Pride, embarrassment	This is the core social loop
Voting	Quick tap vote on phone	Agency, investment	Should feel like applause, not homework
Score reveal	Points, animations, highlights	Competition, joy	Loser needs to feel OK; winner needs to feel great

3.2 Why The Game Is Fun

The answer is always: recognition + surprise + social permission.

Recognition: players see their words woven into a story. The AI used their answer. They matter. Their silly response shaped the comic.

Surprise: the AI takes unexpected directions. The genre the Creative Director chose might be the last thing anyone expected. The narration style transforms mundane answers into dramatic events.

Social permission: the game gives players permission to be absurd, crude, or vulnerable within the safety of a game structure. The AI takes the blame for the story. Players can claim authorship of the funny parts and disavow the embarrassing ones.

3.3 The Recognition Moment

KEY INSIGHT The funniest moment in every round is when a player recognises their own answer inside the AI story. Design the prompt system and LLM instructions to guarantee this moment happens in every single round.

This means: every player answer must appear in the story in a way that is recognisable but transformed. The LLM must be instructed to incorporate every submitted answer, not just the funniest ones. A boring answer made interesting by the AI context is funnier than a funny answer that gets ignored.

4 Prompt System Design

4.1 Design Philosophy

Prompts must provide the LLM with interesting raw material, not blank-filling data. The difference is between "Name a vegetable" (produces: carrot → story says carrot) and "What would you bring to a hostage negotiation?" (produces: a banana → story has to explain why someone brought a banana to a hostage negotiation).

The second type gives the AI something to work with. The prompt forces a contextually inappropriate answer. The context gap is where the comedy lives.

4.2 Prompt Categories

Category	Example Prompt	What It Produces for the LLM
Inappropriate expert	Ce sfat ai da unui chirurg înainte de o operație importantă?	Absurd professional advice in serious context
Social confession	Care e cel mai suspect lucru pe care l-ai făcut de unul singur?	Personal revelation usable as character motivation
Object as character	Dacă telefonul tău ar putea vorbi, ce ar reclama?	Non-human character voice for the story
False expertise	Explică în 10 cuvinte cum funcționează economia mondială.	Overconfident nonsense presented as fact
Relationship accusation	Scrie o acuzație neadevărată despre jucătorul din stânga ta.	Inter-player conflict material for the story
Secret motivation	De ce face cu adevărat [player name] ce face?	Hidden backstory for named character
Crisis response	Ce faci primul când izbucnește un incendiu acasă?	Reveals character priorities under pressure
Romanian specific	Care e scuza ta preferată când întârzi la o nuntă?	Culturally specific, immediately funny to Romanians
Absurd superlative	Care este cel mai periculos lucru din Lidl?	Specific mundane object in dangerous framing
Time pressure	Ai 5 secunde să convingi un vampir să te lase în pace. Ce spui?	Forces impulsive, unconsidered answers

4.3 Prompt Assignment Strategy

Not every player sees the same prompts. The system should:

- Assign 2–3 prompts per player per round
- Ensure no two players answer the exact same prompt (avoids redundant LLM inputs)
- Include at least one prompt that references another named player (generates inter-character dynamics)
- Vary category balance across rounds (no two consecutive rounds with same dominant category)
- Scale prompt difficulty/absurdity with round number (round 1 safe, round 4 chaotic)

4.4 Safe Mode

For family gatherings or younger audiences, a Safe Mode flag filters out:

- Prompts referencing alcohol, relationships, or embarrassment
- LLM instructions for crude humour or adult themes
- Narrator personalities with sarcasm or dark humour

RISK Safe Mode content must be explicitly tagged in the prompt database. Do not rely on LLM filtering at runtime — it is inconsistent. Filter at prompt selection time.

5 Scoring System

5.1 Design Goals

The scoring system must achieve three things simultaneously: reward participation, reward quality, and prevent runaway leaders. A game where one player dominates every round stops being fun. A game where everyone gets equal points removes stakes.

5.2 Point Structure

Action	Points	Notes
Submitting any answer before timer ends	100	Participation baseline — no one scores zero for trying
Your answer appears in the story (confirmed)	200	Guaranteed if LLM is prompted correctly
Voted "funniest panel" (from your answer)	300	Votes come from other players
Voted "best line" (narration referencing your answer)	300	Separate vote category
Voted "most accurate portrayal" of a player	250	For player-targeting prompts
Voting for the eventual winning panel	100	Rewards taste, not just luck
First to submit in round	50	Small speed bonus
Streak: voted best 2 rounds in a row	150 bonus	Comeback mechanic too — streaks reset
Host award (manual, once per game)	200	Host can award for something not captured by voting

5.3 Anti-Runaway Mechanic

After round 2, the player in last place receives a "Scânteia" (spark) modifier: their votes count as 1.5× for one round. This is not announced to them — only the host sees it on the host dashboard. It prevents elimination-feeling without being obvious charity.

DESIGN NOTE The Scânteia must feel like earned luck, not pity. The player should think they just had a great round. Never show the modifier to all players — only the host.

6 The Creative Director

6.1 Concept

The Creative Director (CD) is the brain of the AI pipeline. It is not a genre picker. It is a system that generates a complete creative brief before any story, image, or audio is produced. Every downstream AI component — the LLM, ComfyUI, and ElevenLabs TTS — receives the CD brief and uses it to shape their output.

The result is that each round feels like it was made by a different creative team with a different vision, not like the same template running again.

6.2 Creative Brief Structure

```
CreativeBrief {
  // Narrative
  genre:          string          // e.g. "Telenovelă Românească"
  subgenre:       string          // e.g. "Răzbunarea neașteptată"
  storyStructure: StoryArc        // act structure, beat map
  archetypes:     Archetype[]     // roles player answers will fill
  twists:         Twist[]         // 1-2 mandatory story twists
  comedyLevel:    1-10            // 1=dry/dark, 10=pure slapstick
  toneKeywords:   string[]        // ["melodramatic", "breathless", "ironic"]

  // Presentation
  format:         PresentationFormat // comic | fakeNews | policeReport | documentary
  panelCount:     4 | 5 | 6 | 8
  panelLayout:    LayoutStrategy   // how panels are arranged

  // Visual
  visualStyle:    string          // e.g. "oversaturated Romanian soap opera"
  colourPalette:  string[]        // dominant hex colours for panels
  cameraLanguage: CameraRule[]    // "panel 1 = wide establishing shot"
  lightingMood:   string          // "warm, harsh top light, dramatic shadows"

  // Audio
  narratorPersonality: NarratorPersona
  narratorVoiceKey:  string        // maps to ElevenLabs voice ID (or Piper model ID in
offline mode)
  musicDirection:   string        // "tense strings, sudden silence on twist"
  soundEffects:     SFXNote[]     // per-panel sound effect notes

  // Pacing
  revealPacing:     "slow-burn" | "rapid-fire" | "deliberate" | "chaotic"
  punchlinePanel:   number        // which panel delivers the main joke
}
```

6.3 Genre Registry

Each genre is a complete creative template, not a label. The CD selects one genre per round and populates the brief from that genre's defaults, then randomises within the allowed variance for that genre.

Genre 1: Telenovelă Românească

Attribute	Value
Story structure	Secret revealed → denial → tearful confrontation → unexpected twist → dramatic freeze-frame ending
Character archetypes	Vinovatul (guilty one), Victima (victim), Martorul tăcut (silent witness), Cel care ştia tot (the one who knew everything)
Comedy level	6–8 — melodrama played completely straight is the joke
Visual style	Warm oversaturated colours, extreme close-ups on faces, dramatic zoom panels, tears rendered visibly
Camera language	Panel 1: wide establishing. Panel 2–3: alternating close-ups. Panel 4: shocking revelation. Panel 5: reaction shots. Final: freeze-frame
Narrator personality	Breathless, rhetorical questions, speaks directly to viewer, gasps mid-sentence
Music direction	Dramatic violin, sudden silence before twist, crescendo on final panel
Colour palette	Gold, deep red, amber — warm, passionate, slightly overcooked

Genre 2: Film de Acţiune B Românesc

Attribute	Value
Story structure	Mission briefing → complication → betrayal → improvised revenge → explosion ending
Character archetypes	Eroul ghinionst (unlucky hero), Trădătorul elegant (elegant traitor), Complicele naiv (naive sidekick), Şeful misterios (mysterious boss)
Comedy level	7–9 — sincerity of cheap action cinema applied to absurd situations
Visual style	High contrast, deep shadows, explosion backgrounds, lens flare on every surface
Camera language	Low angle hero shots, extreme close-up on eyes before action, wide for explosions, slow-motion implied in final panel
Narrator personality	Gravelly, terse, uses ellipses heavily, references honour and vengeance often
Music direction	Electric guitar riff, bass drop, dramatic pause before final line
Colour palette	Orange, deep blue, black — classic action poster palette

Genre 3: Basm Românesc Absurd

Attribute	Value
Story structure	Ordinary problem → magical interference → problem gets worse → wise fool solves it accidentally
Character archetypes	Prostul norocos (lucky fool), Înțeleptul inutil (useless wise man), Forța cosmică (cosmic force that could not care less), Animalul filozofic (philosophising animal)
Comedy level	8–10 — pure absurdism, formal fairy-tale register applied to ridiculous events
Visual style	Folk art colour palette, flat perspective, decorative borders inspired by Romanian painted eggs and embroidery
Camera language	Symmetrical compositions, characters always face viewer, no perspective — flat like folk illustration
Narrator personality	Formal fairy-tale register ("A fost odată ca niciodată..."), completely deadpan about impossible events
Music direction	Folk instruments (nai, cobza), whimsical, slightly off-key, melancholy-funny
Colour palette	Deep red, cobalt blue, gold, forest green — Romanian folk art palette

Genre 4: Scandal de Bloc

Attribute	Value
Story structure	Minor grievance → neighbour coalition formation → emergency bloc meeting → absurd resolution that satisfies no one
Character archetypes	Reclamantul profesionist (professional complainer), Acuzatul indignat (indignant accused), Vecinii curioși (nosy neighbours), Administratorul absent (absent building manager)
Comedy level	9 — maximum petty conflict energy, everyone is completely serious about something trivial
Visual style	Cramped panel compositions, many faces crammed into frame, thought bubbles showing private judgements contradicting dialogue
Camera language	Overhead crowd shots, whispering two-shots, dramatic door-opening panels, final panel shows everyone at their own door pretending nothing happened
Narrator personality	Gossipy, conspiratorial, makes knowing asides to reader, takes sides then changes sides

Attribute	Value
Music direction	Tense silence, door slams, muffled arguing from adjacent panels
Colour palette	Institutional beige, harsh fluorescent blue, grey concrete — bloc building palette

Genre 5: Documentar Fals

Attribute	Value
Story structure	Authoritative thesis → contradictory evidence appears → expert panel disagrees with increasing emotion → absurd authoritative conclusion presented as established fact
Character archetypes	Expertul îndoielnic (dubious expert), Subiectul confuz (confused subject), Vocea Oficială (the official voice), Martorul anonim (anonymous witness, face blurred)
Comedy level	5–7 — dry, deadpan, the comedy is how seriously everyone takes nonsense
Visual style	Muted documentary palette, caption boxes with fake credentials, blurred backgrounds, simulated archival footage aesthetic
Camera language	Talking-head interview panels, b-roll cutaway panels, caption-heavy layouts, one panel must be "security camera footage"
Narrator personality	BBC documentary register, measured and authoritative, never acknowledges the absurdity, treats everything as established fact
Music direction	Ambient drone, occasional tense musical sting, long silences
Colour palette	Desaturated grey-green, off-white, institutional blue — documentary television palette

Genre 6: Horror Mioritic

Attribute	Value
Story structure	Omen ignored → escalation of signs → confrontation with inevitable force → fatalistic acceptance and poetic ending
Character archetypes	Cel marcat (the marked one), Sfătuitorul ignorat (the ignored advisor), Forța naturii (the natural force — not necessarily evil), Animalul care știe (the knowing animal)

Attribute	Value
Comedy level	3–5 — the comedy is the Romanian cultural attitude of calm acceptance toward disaster, not the disaster itself
Visual style	Desaturated, fog effects, silhouettes against sky, woodcut-inspired line work, wolves optional but appreciated
Camera language	Atmospheric wide shots, long silences implied by empty panels, face never fully visible until final panel
Narrator personality	Fatalistic, poetic, uses metaphor from nature, completely calm about terrible things, occasionally philosophical
Music direction	Solo doina, wind sounds, distant bells, silence at moment of revelation
Colour palette	Grey-blue, forest green, charcoal, single gold accent — Carpathian palette

Genre 7: Știri Rupte din Realitate

Attribute	Value
Story structure	Breaking news intro → contradictory reporter on the ground → social media eruption → expert analysis that misses the point → resolution that raises more questions
Character archetypes	Prezentatorul de știri (news anchor), Reporterul confuz (confused field reporter), Expertul de serviciu (on-call expert), Cetățeanul revoltat (outraged citizen)
Comedy level	8–9 — satirises Romanian news media specifically, recognisable to all Romanian players
Visual style	News broadcast aesthetic, lower-third graphic boxes, split-screen panels, ticker tape text, aggressive red colour scheme
Camera language	Talking-head panels with graphic overlays, chaotic split-screen, one panel must be a phone-filmed vertical video
Narrator personality	Breathless news anchor energy, treats everything as the biggest story of the century, dramatic pause before every number
Music direction	News jingle stab, urgent music bed, social media notification sounds
Colour palette	Aggressive red, white, black — Romanian television news palette

6.4 Presentation Formats

The Creative Director also selects a presentation format. Genre determines the story. Format determines how the story is visually presented. These are independent axes.

Format	Description	Compatible Genres
Western Comic	Standard panel layout, speech bubbles, action lines	All genres
Fake News Broadcast	TV news aesthetic, lower thirds, graphics	Știri Rupte, Documentar Fals
Police Report	Bureaucratic document with hand-drawn witness sketches	Scandal de Bloc, Acțiune B
Documentary Film	Mixed panel/caption layout, interview stills, b-roll	Documentar Fals, Horror Mioritic
Folk Tale Illustration	Single-page illustrated spread, flat folk art style	Basme Românești
Instagram Story Sequence	Vertical panels, phone-native aesthetic, emoji overlays	Știri Rupte, Telenovelă
Interpol Dossier	Case file aesthetic, redacted text, surveillance photos	Acțiune B, Documentar Fals

7 AI Pipeline Architecture

7.1 Provider Interface Design

Every AI component is accessed through a provider interface. The game logic never calls Ollama, FLUX, ElevenLabs, or Piper directly. It calls the interface. The interface has concrete implementations for each provider, and the default can be swapped at any time.

This means: if a better LLM is released tomorrow, we write a new implementation of StoryLLMProvider and swap it in. Zero changes to game logic.

```
// Provider Interfaces (Python abstract base classes)

class StoryLLMProvider(ABC):
    @abstractmethod
    def generate_story(self, brief: CreativeBrief, answers: PlayerAnswers) -> Story: ...

class ImageGeneratorProvider(ABC):
    @abstractmethod
    def generate_panel(self, prompt: ImagePrompt, style: VisualStyle) -> PanelImage: ...

class TranslatorProvider(ABC):
    @abstractmethod
    def translate(self, text: str, source: str, target: str) -> str: ...

class TTSPProvider(ABC):
    @abstractmethod
    def synthesise(self, text: str, persona: NarratorPersona) -> AudioFile: ...

// Concrete implementations
class OllamaStoryLLM(StoryLLMProvider): ...
class OpenAISToryLLM(StoryLLMProvider): ...
class FluxImageGenerator(ImageGeneratorProvider): ...
class OpenAIImageGenerator(ImageGeneratorProvider): ...
class ElevenLabsTTS(TTSPProvider): ...
class PiperTTS(TTSPProvider): ...
```

7.2 Pipeline Execution Flow

PIPELINE ORCHESTRATOR

Input: PlayerAnswers, GameState

```

↓
[1] CREATIVE DIRECTOR
    → Selects genre (weighted random, avoids recent)
    → Generates CreativeBrief (full creative specification)
    → Assigns player answer → archetype mappings
    ↓
[2] STORY LLM (Ollama + Llama 3.1 8B)
    Input: CreativeBrief + PlayerAnswers
    Output: Story struct {
        title: string (Romanian)
        panels: PanelDescription[] (Romanian)
        narratorScript: string[] (Romanian, per panel)
        imagePrompts: string[] (English — translated internally)
    }

```

```

VRAM: ~4-6GB | Time: ~15-25s
  ↓
  ▼ [VRAM CLEARED]
[3] IMAGE GENERATOR (ComfyUI + FLUX.1 schnell)
Input:  imagePrompts[] + VisualStyle + CharacterDescriptions
Output: panel_1.png ... panel_N.png
VRAM: ~10-11GB | Time: ~8-15s per panel
Total: ~50-90s for 6 panels
  ↓
  ▼ [VRAM CLEARED]
[4] TTS (ElevenLabs default / Piper offline fallback)
Input:  narratorScript[] + NarratorPersona
Output: narration_1.wav ... narration_N.wav
VRAM: minimal (CPU) | Time: ~1-3s per panel (ElevenLabs API) / ~3-5s (Piper fallback)
  ↓
[5] ASSET PACKAGER
Output: round_XXX/ {
        panels/, narration/, story.json,
        brief.json, metadata.json
      }
Signals Node.js server: pipeline_complete

```

7.3 Character Consistency Strategy

Strict face-identity locking across AI panels is an unsolved problem in open-source image generation. The approach here is visual consistency through description constraints, not biometric identity.

Technique	How It Works	Effectiveness
Character Description Sheet	Each character assigned at story-gen time: hair colour, clothing, distinguishing feature. Same description in every panel prompt.	High — consistent silhouette and colour
Colour-coded clothing	Each player-character gets a unique dominant clothing colour. Reader identifies character by colour, not face.	High — instinctive visual parsing
Visual style lock	Every panel gets the same LoRA/style tokens. Face variation is hidden by consistent art style.	Medium — depends on style strength
Panel composition rules	CD specifies which character occupies which panel region. Prevents accidental character confusion.	Medium — requires reliable prompt following
Character count limit	Maximum 3 characters per panel. More than 3 and consistency collapses.	High — hard constraint enforced in orchestrator

7.4 Romanian → English Translation Layer

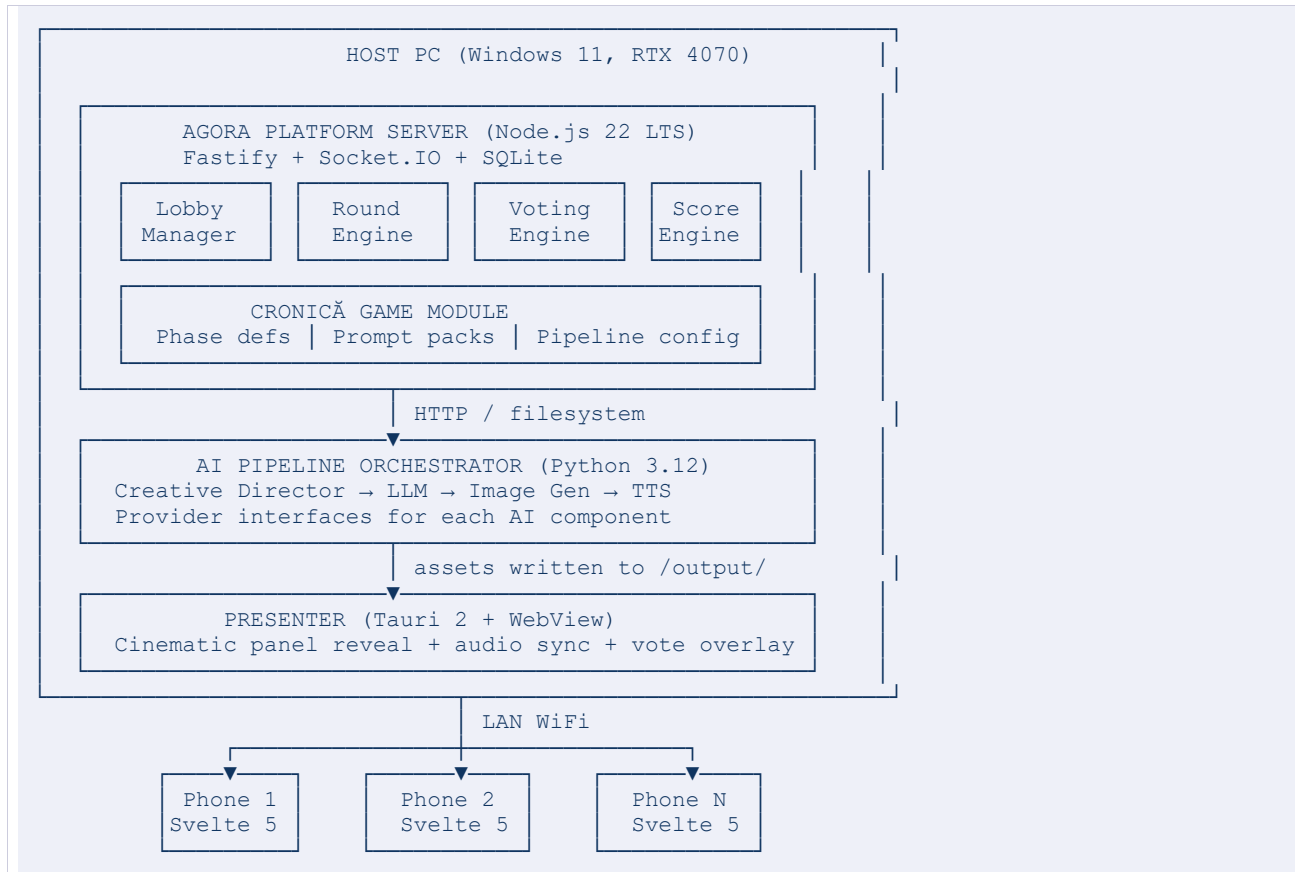
FLUX.1 and all major image generation models were trained predominantly on English text. Romanian prompts produce degraded results. The pipeline maintains a strict separation:

- Story, narrator scripts, panel descriptions: Romanian throughout
- Image generation prompts: translated to English before hitting ComfyUI
- Translation is performed by the LLM itself in the story generation step — not a separate translation call
- The LLM is instructed to output image prompts in English as part of its structured JSON response

IMPLEMENTATION NOTE Instruct the LLM to output a JSON struct with Romanian fields and English imagePrompt fields in a single call. Avoids a separate translation API call and keeps the image prompt contextually grounded in the story the LLM just wrote.

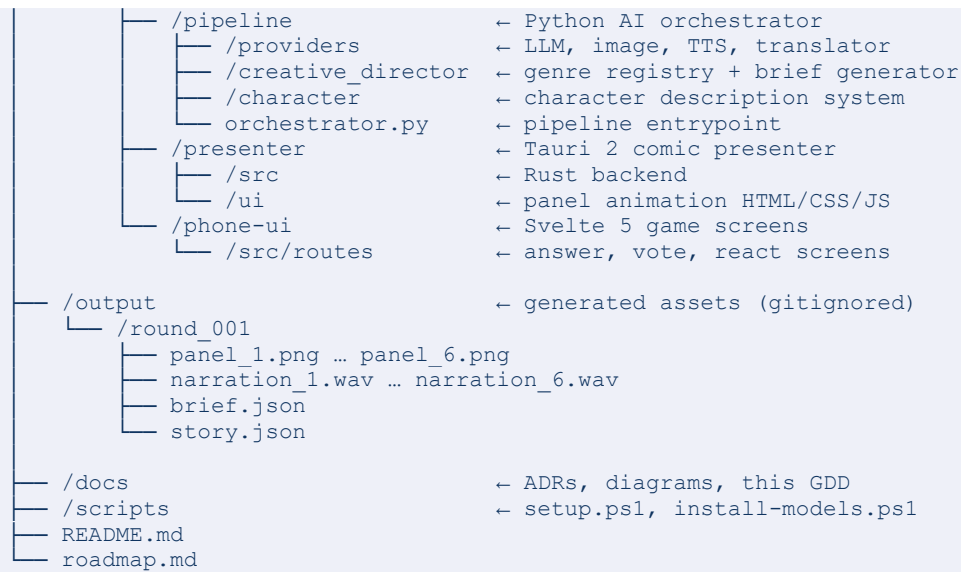
8 Technical Architecture

8.1 System Overview



8.2 Folder Structure



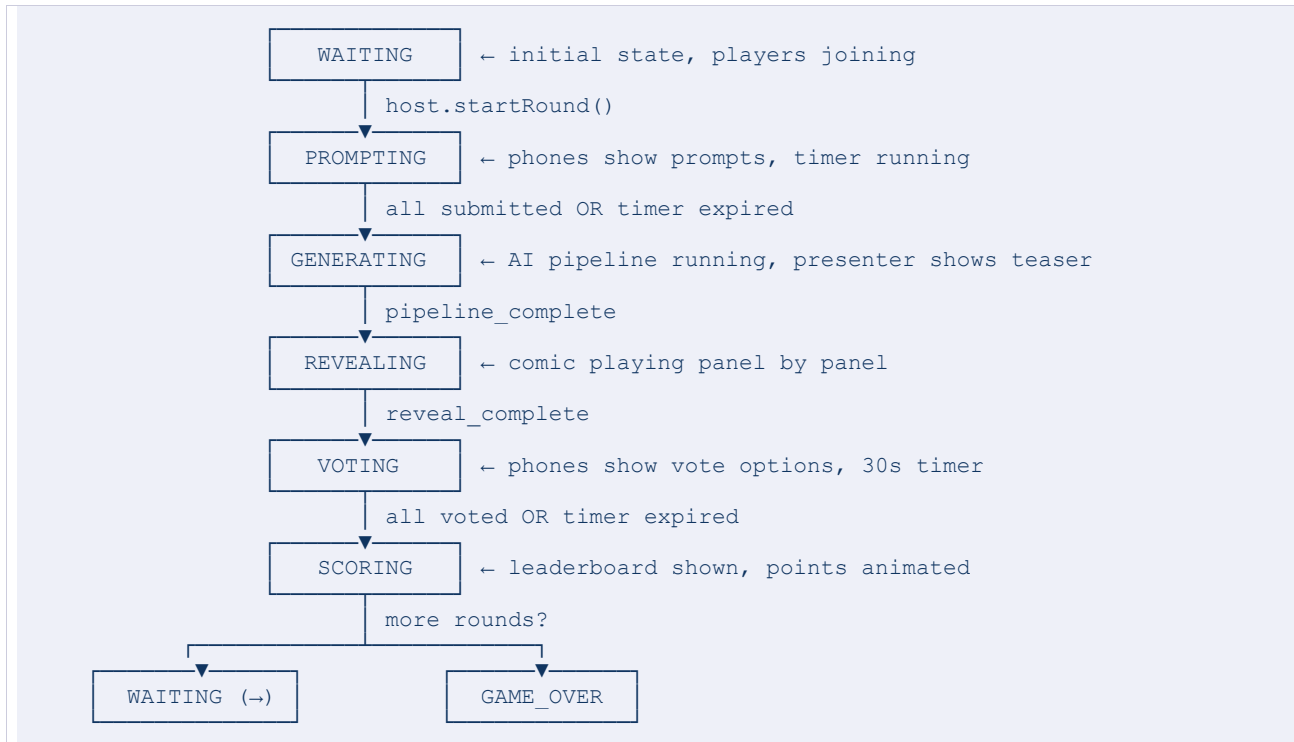


8.3 Data Flow: One Complete Round

EVENT	ACTOR	DATA
host_start_round	Node Server	→ emits prompts to each phone via Socket.IO
player_submit_answer	Socket.IO	→ stored in SQLite round_answers table
all_players_submitted	Round Engine	→ triggers pipeline via HTTP POST /pipeline/run
pipeline_run	Python Orch.	→ reads answers, selects genre, runs CD
creative_brief_ready	Creative Director	→ writes brief.json to /output/round_XXX/
story_ready	Ollama LLM	→ writes story.json with RO text + EN prompts
panels_ready	ComfyUI/FLUX	→ writes panel_N.png files
narration_ready	ElevenLabs/Piper	→ writes narration_N.wav files
pipeline_complete	Python Orch.	→ POST /pipeline/complete to Node server
trigger_presenter	Node Server	→ IPC message to Tauri window
reveal_starts	Tauri Presenter	→ begins cinematic panel sequence
vote_phase_starts	Node Server	→ emits vote options to all phones
player_vote	Socket.IO	→ stored in SQLite votes table
scoring_complete	Scoring Engine	→ emits leaderboard update to all phones
round_end	Round Engine	→ state reset, await next round

9 Component & State Diagrams

9.1 Round State Machine



9.2 Socket.IO Event Reference

Event	Direction	Payload	Phase
player:join	Client → Server	{ nickname, roomCode }	WAITING
player:joined	Server → All	{ players[] }	WAITING
round:start	Server → All	{ roundNumber, prompts[] }	PROMPTING
player:submit	Client → Server	{ answers[] }	PROMPTING
round:generating	Server → All	{ genreTitle, estimatedSeconds }	GENERATING
round:reveal_ready	Server → Presenter	{ assetPath }	GENERATING
round:reveal_panel	Server → All	{ panelIndex }	REVEALING

Event	Direction	Payload	Phase
round:vote_start	Server → All	{ voteOptions[] }	VOTING
player:vote	Client → Server	{ voteType, targetId }	VOTING
round:scores	Server → All	{ scores[], delta[] }	SCORING
player:react	Client → Server	{ emoji }	REVEALING
player:disconnect	Auto	{ playerId }	Any

10 Technical Risks & Mitigations

Risk	Severity	Probability	Mitigation
VRAM overflow: LLM + image gen simultaneously exceed 12GB	HIGH	HIGH	Sequential pipeline with explicit VRAM clear between steps. Monitor with nvidia-smi. Hard limit: never run two AI processes simultaneously.
LLM produces Mad Lib output despite instructions	HIGH	MEDIUM	Extensive prompt engineering. Eval suite to test story quality before ship. Genre brief injection in system prompt, not user prompt.
Character inconsistency breaks immersion	MEDIUM	HIGH	Adopt colour-coded character design. Accept variation as comic art style. Never promise face identity.
Phone browser incompatibility (older Android)	MEDIUM	MEDIUM	Target ES2020. Test on Android 10+, iOS 14+. No Web APIs beyond fetch and WebSocket.
FLUX generates inappropriate content	HIGH	LOW	Negative prompts always active. Content filter wrapper on all image outputs. Safe Mode blocks flagged prompts at source.
ElevenLabs API unavailable or rate-limited during play session	MEDIUM	LOW	Piper TTS is bundled as offline fallback. Pipeline auto-detects API failure and switches. Piper quality is robotic but functional.
Generation takes >3 min and kills energy	HIGH	MEDIUM	Cinematic loading experience with teaser reveals. Genre title reveal at start of generation. Real-time phone reactions during wait.
LLM hallucinates player names or ignores answers	HIGH	MEDIUM	Structured JSON output enforced. Post-generation validation that all player names and at least N answers appear in story.
Local network discovery fails on some routers	MEDIUM	MEDIUM	Display IP address as manual fallback alongside QR code. mDNS for auto-discovery.
FLUX.1 schnell license changes	HIGH	LOW	Provider interface means swap is one class replacement. Monitor Black Forest Labs announcements.

11 Milestones & Complexity

11.1 Milestone Overview

Milestone	Name	Deliverable	Complexity	Est. Duration
M0	GDD Approval	This document approved, architecture finalised	Low	1–2 days
M1	Platform Skeleton	Node server + Socket.IO lobby + phone shell + round state machine	High	5–7 days
M2	Prompt & Answer Loop	Full PROMPTING phase: prompts delivered, answers collected, timer works	Medium	3–4 days
M3	Creative Director	Genre registry, brief generation, archetype assignment	High	4–5 days
M4	Story Generation	Ollama + Llama 3.1 8B integration, structured story output, RO→EN translation	High	5–7 days
M5	Image Pipeline	ComfyUI + FLUX.1 schnell, character descriptions, panel output	Very High	7–10 days
M6	TTS Pipeline	ElevenLabs integration (default), Piper fallback, narrator persona mapping, audio output	Medium	3–4 days
M7	Presenter	Tauri window, panel animation, audio sync, cinematic reveal	High	6–8 days
M8	Full Game Loop	Voting, scoring, Scânteia mechanic, leaderboard, multi-round	High	5–6 days
M9	Polish & UX	Romanian UI copy, loading experience, sound effects, reactions	Medium	4–5 days
M10	Playtest Build	Complete game, installable, tested with real players	Medium	3–4 days

Milestone	Name	Deliverable	Complexity	Est. Duration
M11	Distribution	Windows installer, model download script, README, public build	Medium	3–4 days

11.2 M1 Detailed Breakdown (First Milestone)

M1 is the most critical milestone because every subsequent milestone builds on it. It must be engineered correctly. Speed is not the goal.

Task	Description	Done When
Monorepo setup	Git repo, folder structure, tsconfig, shared types	npm run build works from root
Node server bootstrap	Fastify + Socket.IO, basic health endpoint, env config	Server starts, /health returns 200
Room management	Create room, generate code, QR code generation, room state	Host creates room, QR appears
Player join flow	Phone opens URL, enters name, appears on host screen	Player visible on host within 2s of scan
Connection resilience	Reconnection logic, player state preserved on disconnect	Phone can reconnect without losing state
Round state machine	State transitions, phase sequencing, event emissions	State transitions logged correctly
Phone shell (Svelte)	Join screen, name entry, wait screen, connection indicator	Phone UI renders on Android + iOS
SQLite schema	Rooms, players, rounds, answers, votes tables	Schema migration runs cleanly
Integration test	Simulate 4 players joining, round starting, state advancing	All 4 phones see correct state

12 Future Expansion

12.1 Future Games on AGORA Platform

Game Concept	Mechanic	AI Component Needed
Judecata Poporului	Players debate absurd court cases, AI plays judge	LLM judge persona
Știri False	Players write fake headlines, AI generates fake article + image	LLM + image gen
Karaoke de Poveste	AI writes a song in a genre using player inputs, players perform	LLM + music gen (future)
Profeția Bătrânei	Players describe events, AI writes a prophecy, group interprets	LLM only
Ghici Cine	AI generates anonymous character descriptions, players guess whose	LLM + voting
Filmul Noaptea	AI generates a B-movie script using player inputs, players perform scenes	LLM + TTS

12.2 Platform Expansion Features

- Streamer mode: OBS overlay output, Twitch chat integration, viewer voting
- Spectator mode: additional browsers join as viewers without playing
- Replay system: save and replay best rounds, shareable as video
- Prompt editor: host can create custom prompt packs
- Cloud mode: optional cloud AI provider for faster generation
- Localisation: English, French, Hungarian (Romanian minority language) support
- Mobile app: optional native wrapper for improved phone experience
- Analytics dashboard: track funniest prompts, most popular genres

12.3 AI Pipeline Evolution Path

Phase	LLM	Image Gen	TTS	Trigger
Launch (local)	Llama 3.1 8B via Ollama	FLUX.1 schnell via ComfyUI	ElevenLabs Romanian (default), Piper (offline fallback)	Now
Quality upgrade	Llama 3.3 70B (if VRAM allows) or cloud	FLUX.1 dev (non-commercial) or SDXL Lightning	ElevenLabs (optional)	Better hardware or cloud mode
Commercial cloud	Claude claude-sonnet-4-6 or GPT-4o	Replicate / Fal.ai FLUX	ElevenLabs Romanian	Commercial launch decision
Multimodal	GPT-4o vision for prompt reaction	Video generation (Sora, Wan)	Voice cloning	2026+

13 Open Decisions Requiring Approval

The following decisions are documented here because they affect architecture. They do not need to be resolved before M1, but they should be resolved before the milestone they impact.

Decision	Options	Impact	Needed Before
Monetisation model	Free / paid / freemium	AI model licence choices, distribution strategy	M10
Distribution channel	itch.io / Steam / direct installer / open source	Installer format, analytics, update mechanism	M11
Cloud AI fallback	Yes (OpenAI/Replicate) / No (local only)	Provider interface already supports it — just needs keys	M5
Multiplayer over internet	LAN only / NAT traversal / hosted relay	Significant networking complexity if yes	M8
Save/replay system	Yes / No	Output folder structure, storage budget	M7
Streamer mode priority	M9 / post-launch	Affects Tauri presenter output format	M9
Romanian TTS voice	DECIDED: ElevenLabs as default, Piper as offline fallback	ElevenLabs API key required for default mode. Piper bundled for offline LAN play.	RESOLVED

14 Glossary

Term	Definition
AGORA	The reusable party game platform. All games run on AGORA.
CRONICĂ	The first game built on AGORA. An AI comic generator party game.
Creative Director (CD)	The system component that generates the complete creative brief for each round.
Creative Brief	The complete creative specification produced by the CD: genre, visual style, narrator persona, panel count, etc.
Provider Interface	An abstract interface for an AI component (LLM, image gen, TTS, translator). Implementations are swappable.
Game Module Interface	The contract a game must implement to run on the AGORA platform.
Scânteia	The anti-runaway scoring mechanic. Last-place player gets hidden vote multiplier.
Pipeline	The sequential AI generation process: Creative Director → Story LLM → Image Gen → TTS.
Presenter	The Tauri 2 desktop window responsible for the cinematic comic reveal.
Phone Shell	The shared Svelte 5 base UI that all AGORA games use for joining and navigation.
Recognition Moment	The moment when a player sees their answer incorporated into the AI story. The core social loop.
Panel	One frame of the comic. A round produces 4–8 panels depending on the Creative Director's decision.
Archetype	The narrative role assigned to a player's answers within the genre structure.

15 Approval & Next Steps

This document is ready for review. Before Milestone 1 begins, the following must be confirmed:

1. Architecture approval: AGORA platform design, monorepo structure, and component boundaries
2. Tech stack confirmation: Node.js 22 + Fastify + Socket.IO, Svelte 5, Tauri 2, Python orchestrator
3. AI stack confirmation: Ollama + Llama 3.1 8B, ComfyUI + FLUX.1 schnell, ElevenLabs TTS (Piper offline fallback)
4. Creative Director design: genre registry (7 genres), presentation formats, brief structure
5. Prompt system design: category types, assignment strategy, Safe Mode requirement
6. Scoring system: point structure, Scânteia mechanic approved
7. Provider interface pattern: all AI components behind abstract interfaces

READY TO BUILD Once this GDD is approved, development begins at M1: Platform Skeleton. Estimated first playable prototype: M5 completion (approximately 4–5 weeks of focused development).